

6 The mammalian kidney is responsible for:

- the excretion of urea
- osmoregulation (the homeostatic control of the water potential of the blood).

(a) (i) Outline how **and** where the excretory product urea is made in the body.

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(ii) Homeostatic control of the water potential of blood includes receptors, effectors and target cells.

Identify the names **and** locations of these components of homeostatic control in osmoregulation.

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(b) The glomerulus and Bowman's capsule of the nephron are important in the formation of urine.

Outline the role of the glomerulus and Bowman's capsule in the formation of urine.

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- (c) Concentrated urine contains a high concentration of solutes and a small volume of water. Different species of mammals vary in their ability to produce urine with a high solute concentration.

Table 6.1 compares the ratio of the solute concentration of urine (U) to the solute concentration of blood plasma (P) in some mammal species. The habitats of the mammal species are also shown.

Table 6.1

mammal species	maximum ratio of solute concentration of urine to solute concentration of blood plasma (U : P)	habitat
beaver	1.7 : 1	rivers and lakes
human	4.5 : 1	variable
camel	8.0 : 1	desert
rat	9.0 : 1	variable
kangaroo rat	16.0 : 1	desert

With reference to Table 6.1, suggest what the different values of U : P show about the ability of these mammal species to tolerate a shortage of water in their environment.

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